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mrtwinklesharma



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Electronic Mail is a widely used method to transfer/exchange messages among people. We generally send the email with any software or application like Gmail, outlook, thunderbird mail, and yahoo, etc. We can also code the basic concept of these applications in a node.js app by using any third-party library which can interact with the networking systems and send an email.

Nodemailer: There are various modules available for sending emails but the nodemailer is the most popular one and it provides us simple procedure and functionality to send mail.

Features of Nodemailer:

- It supports various features like adding HTML in the mail, Unicode characters, sending attachments with the text, etc.
- It uses the simple mail transfer protocol (SMTP).
Below is the step-by-step approach to be followed to integrate this module into our application.

Step 1: Module Installation: Write the command in the terminal to install nodemailer and then import at the top of your nodejs application.



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Third party
module
Installation

```
npm install nodemailer
```

Now we are ready to import this into our application.

```
const nodemailer =  
require('nodemailer');
```

Step 2: Create Transporter Object: There exists a createTransport method in nodemailer which accepts an object with some configurations and finally returns a transporter object. This object will be required later to send emails.

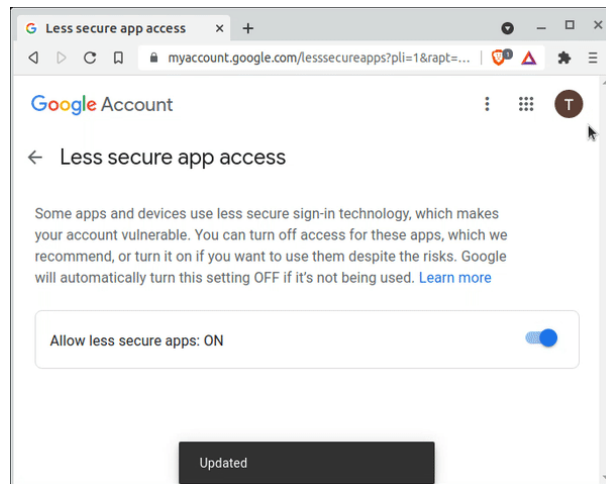
It is used
for sending
mail.

```
const transporter =  
nodemailer.createTransport(transport[  
, defaults]);
```

method accepts
an object with
configuration
settings &
returns
Transporter Obj.

Here we are using Gmail as a service just for sample purposes, Although nodemailer can be easily integrated with any other mail service. In Gmail, we can either make our account less secure or we can use the Oauth2 security authentication unless normally google will not allow sending any mail via node.js.

Less Secure Account: Visit this link to make your account less secure, after doing this we can create our working transporter object just with the username and password of your Gmail account.



app.js

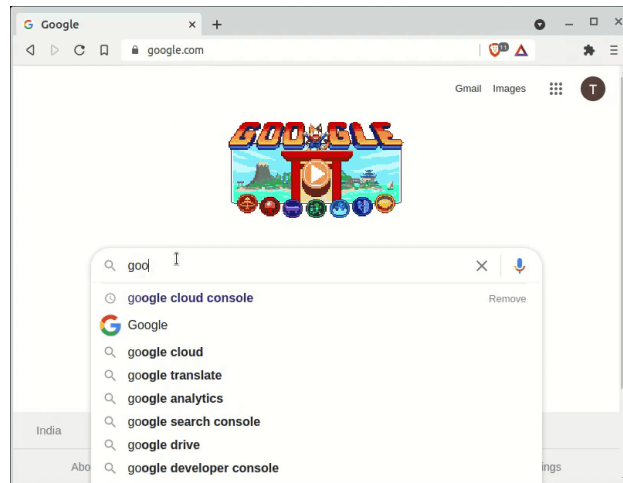
```
const nodemailer = require('nodemailer')

const transporter = nodemailer.createTr
  service: 'gmail',
  auth: {
    user: secure_configuration.EMAIL_US
    pass: secure_configuration.PASSWORD
  }
}) ;
```

Using Oauth2: According to [official docs](#) here we need to provide client id, client secret, refresh token, and an access token along with the username and password. Follow the step-by-step approach to get these configurations from the google cloud console.

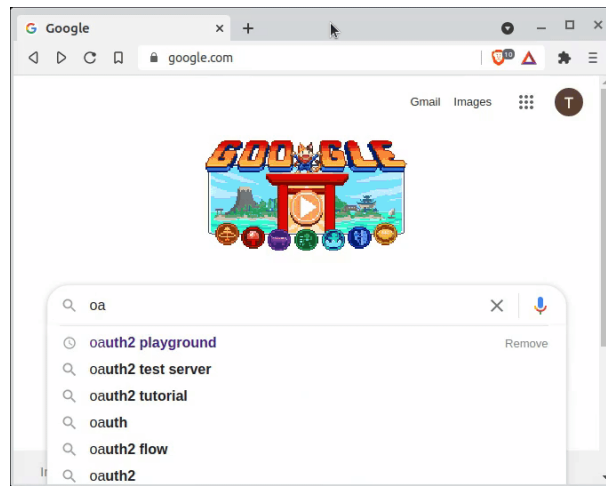
1. Open Google Cloud Console: In this step, we will get our client id and client secret. Visit the google cloud console website and register/sign in yourself. Then proceed to **API & Services** section from the leftmost navigation bar. Now check out the dashboard and create a project. After this, visit the Oauth consent screen to register about your application in this step make sure to select user type **external** and add some/one **Test Users**. After this step, go to the credentials section and click on Create credentials and then choose Oauth2 ClientID and choose application type as a web application, also make sure to add **redirect URI as OAuth playground** ([copy link from here](#)). Finally, you'll be successful to get your **client_id** and

client_secret.



2. Open the OAuth2 Playground: Here we will get our refresh token and access token. Visit the [OAuth2 playground](#). Click on the oauth2.0 configuration icon from the right and then check the **use your own credentials** checkbox and provide the same client id and secret you've got from the cloud console. Now select the Gmail API for authorization. Click on Authorize API and then authorize it by the same Gmail id you have been filled as a test user in the credentials section of the last step. Finally, click on the exchange authorization code for tokens and this will provide a **refresh token and access token**.





Note: We suggest you open these gifs in a separate tab and follow both approaches separately and don't skip at any span this would save you from a lot of confusion, Also don't try to copy this client id and client secret, etc. it will not going to work.

app.js

```
const nodemailer = require('nodemailer')

const transporter = nodemailer.createTransport({
  service: 'gmail',
  auth: {
    type: 'OAuth2',
    user: secure_configuration.EMAIL_US,
    pass: secure_configuration.PASSWORD,
    clientId: secure_configuration.CLIENT_ID,
    clientSecret: secure_configuration.CLIENT_SECRET,
    refreshToken: secure_configuration.REFRESH_TOKEN,
  }
})
```

With this our transporter object is ready and now we can send our emails.

Step 3: Configure eMail: Before sending the mail we have to create some message configurations like what to send where to send etc. It is super easy to create these configurations there are several [key-value pairs](#), from which you can provide the required ones along with some other values to the predefined key.

Sending a simple text to one email:

Javascript

```

const mailConfigurations = {
  // It should be a string of sender
  from: 'mrtwinklesharma@gmail.com',

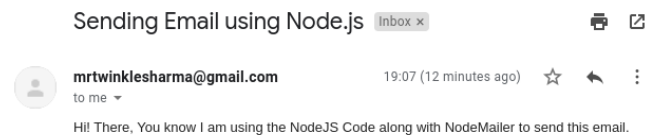
  // Comma Separated list of mails
  to: 'smtwinkle451@gmail.com',

  // Subject of Email
  subject: 'Sending Email using Node.',

  // This would be the text of email
  text: 'Hi! There, You know I am using
    + ' NodeJS Code along with NodeMailer
    + ' to send this email.'
};

```

Output: If we send the email with these configurations, something like this will be sent to the receiver. Although our code is currently incomplete, here we are just showing you how this message configuration will look like when the code will be completed.



Sending to Multiple emails: We can concatenate more emails with a comma as a separator.

Javascript

```

const mailConfigurations = {
  from: 'mrtwinklesharma@gmail.com',
  to: 'smtwinkle451@gmail.com, anyoth
  subject: 'Sending Email using Node
  text: 'Hi! There, You know I am using
    + ' along with NodeMailer to sen
};

```

Sending some HTML content: Just replace the

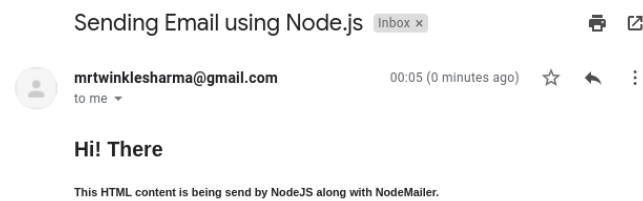
Plain text with HTML and provide it to *html* key.

Javascript



```
const mailConfigurations = {  
  from: 'mrtwinklesharma@gmail.com',  
  to: 'smtwinkle451@gmail.com',  
  subject: 'Sending Email using Node.js',  
  html: "<h2>Hi! There</h2> <h5> This H  
        being send by NodeJS along with No  
};
```

Output:



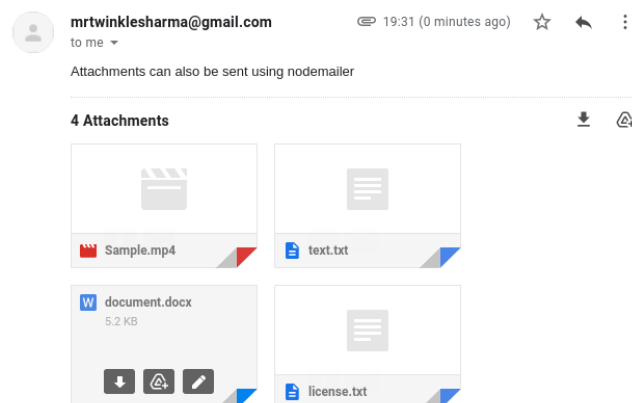
- **Sending some Attachments**, Nodemailer is so flexible while sending attachments, you can send any type of file which is being accepted by email service.

Javascript




```
const mailConfigurations = {
  from: 'mrtwinklesharma@gmail.com',
  to: 'smtwinkle451@gmail.com',
  subject: 'Sending Email using Node.js',
  text: 'Attachments can also be sent us',
  attachments: [
    {
      // utf-8 string as an attachment
      filename: 'text.txt',
      content: 'Hello, GeeksforGeeks Lear',
    },
    {
      // filename and content type is der
      path: '/home/mrtwinklesharma/Progra',
    },
    {
      path: '/home/mrtwinklesharma/Videos',
    },
    {
      // use URL as an attachment
      filename: 'license.txt',
      path: 'https://raw.githubusercontent.com/nodem',
    }
  ]
};
```

Output:



Not only these but nodemailer have a lot of possibilities for message configurations you can check them all from [here](#).

Step 4: Send eMail: Use any one of the above approaches to proceed with the transporter object, and then choose any one email configurations to send mail.

There exists a **sendMail** method in the transporter object which accepts email configurations and a callback function that will be executed either when

mail has been sent or due to an error.

```
transporter.sendMail(mailConfigurations[, callback]);
```

Javascript



```
transporter.sendMail(mailConfigurations
    if (error) throw Error(error);
    console.log('Email Sent Successf
    console.log(info);
});
```

Step 5: Combine them All

You can pick any options from the step 2 and 3.

After providing it to the sendMail method you will be successfully able to send an email with node.js.

Explanation: Here we have imported the nodemailer module in the beginning and then used the Oauth2 type of authentication, later there is the most basic message configuration which is being used. Finally, the **sendMail** method is sending the mail to receivers provided in message configuration.

Note:- The secure module being imported has nothing to do with this nodemailer, I have just used it to secure my credentials.

app.js



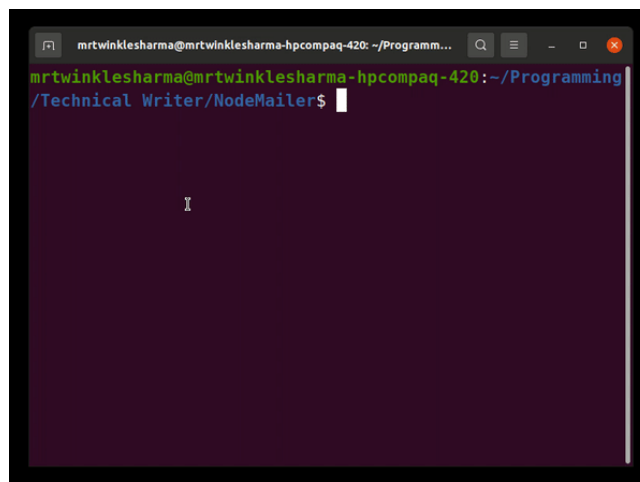
```
const nodemailer = require('nodemailer')
const secure_configuration = require('..')

const transporter = nodemailer.createTransport({
  service: 'gmail',
  auth: {
    type: 'OAuth2',
    user: secure_configuration.EMAIL_USER,
    pass: secure_configuration.PASSWORD,
    clientId: secure_configuration.CLIENT_ID,
    clientSecret: secure_configuration.CLIENT_SECRET,
    refreshToken: secure_configuration.REFRESH_TOKEN
  }
})

const mailConfigurations = {
  from: 'mrtwinklesharma@gmail.com',
  to: 'smtwinkle451@gmail.com',
  subject: 'Sending Email using Node.js',
  text: 'Hi! There, You know I am using Node.js'
}

transporter.sendMail(mailConfigurations, (error) => {
  if (error) throw Error(error)
  console.log('Email Sent Successfully')
  console.log(info)
})
```

Output: Run this code snippet with node command and this will be the output in the console and Gmail inbox.



Sending Email using Node.js Inbox x



mrtwinklesharma@gmail.com
to me ▾

19:07 (12 minutes ago) ☆ ↩ ⋮

Hi! There, You know I am using the NodeJS Code along with NodeMailer to send this email.

Last Updated : 11 Sep, 2021



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